Assignment

ESSAY GRADE

- A student was interested in whether there was a positive relationship between the time spent doing an essay and the mark received.
- He got 45 of his friends and timed how long they spent writing an essay (**hours**) and the percentage they got in the essay (**essay**).
- He translated these grades into their degree classifications (**grade**).

Load:
EssayMarks.dat

- Find out the correlation between the time spent on an essay and the eventual grade mark in class.

Assignment

CHECK THE PAPER

Think Hard or Think Smart: Network Reconfigurations After Divergent Thinking Associate With Creativity Performance

Load: **CAQ_orig_flu.dat**

Creativity Performances: CAQ and AUT

The average CAQ score for the 40 participants was 10.8 ± 10.4, ranging from 1 to 62. Regarding the overall AUT performances, the evaluated originality and fluency were 0.33 ± 0.11 and 3.1 ± 1.0, respectively. The CAQ scores were adjusted by the log-transform (Form and Kaernbach, 2018), and the log(CAQ) scores exhibited a positive trend toward the AUT fluency ($r = 0.30$, $p = 0.06$), without prominent relationships with originality ($r = 0.22$, $p = 0.16$) for all participants. To test the difference in brain activity between high and low creativity individuals, the following results were divided into two groups with equal sample size ($n = 20$ for each group); where their CAQ scores were 16.6 ± 12.2 for the high-creative group and 5.0 ± 1.9 for the low-creative group. In addition, the two groups showed no significant differences in age ($p = 0.06$), gender ($p = 0.53$) and motion during the two rs-fMRI scans (framewise displacement: $p = 0.39$).

- **Check the 1st paragraph of the Results section in the paper.**
- **Please reproduce the findings in the first paragraph of Results section, with given dataset.**
- **Plot the correlation matrix, only displays the cell with significance.**

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CAQ: creativity achievement questionnaire
AUT: alternative usage task for creativity
Originality & **Fluency**: AUT performance

Assignment

SUPERMODEL

- A fashion student was interested in factors that predicted the salaries of catalog models.
- She collected data from 231 models. For each model she asked them their gross income (**salary**), their age (**age**), how many years they had worked as a model (**years**) and then got a panel of experts from modelling agencies to rate the attractiveness of each model as a percentage index (**beauty**).

Load:
Supermodel.dat

- If you were this fashion student, how would you do the regression analysis to check which variables better predict a model's salary?
- How valid is this model? Please explain your rationale to justify the model you provide.

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Assignment

CHILD AGGRESSION

- A psychologist planned to explore the relationship between aggression and several potential predicting factors in 666 children who had an older sibling.
- He designed to measure following indices:
  - **Parent Style** (high = bad parenting practices),
  - **Television** (high = good diet low in additives),
  - **Diet** (high = more time spent watching TV),
  - **Computer** Games (high = more time spent playing computer games),
  - **Sibling Aggression** (high = more aggression in older sibling).

Load:
ChildAggression.dat

- Past research indicated that parroting style and sibling aggression were good predictors of the level of aggression in the younger child.
- Can you prove that statement through the factors? or new hobbies affect their aggression as well? Please confirm the speculation.

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